

KhetiMitra

Intelligent Decision Support for India's 140 Million Farmers

₹63K Cr

Lost Annually

140M

Farmers Targeted

86%

Underserved

Team: Bharat AgriNauts

Hackathon Submission :- Sarang Ahlawat • Akshat Raheja • Akhil Sharma



THE PROBLEM

India's Farmers Are Flying Blind

₹63,000
CRORE

lost by Indian farmers annually due to
poor crop decisions & climate shocks



86% Small & Marginal

No access to expert advisory or real-time data



Zero Digital Support

Farmers rely on word-of-mouth and seasonal luck



Policy Blindspot

Crores in govt schemes go unclaimed due to unawareness

700 million rural Indians. One broken information pipeline.

A Trillion-Rupee Gap, Waiting to Be Bridged

140M

Total Addressable

Indian farmers

28M

Serviceable Market

Smartphone-enabled farmers

4.2M

Beachhead Pilot

Haryana & Punjab farmers

 **Haryana Pilot:** 3 districts • 3 crop seasons • 7 farmer profiles • Extendable to 15 Indian states

India's AgriTech sector is projected to exceed ₹70,000 Crore by 2025 - with advisory services as the largest under-served segment.

THE SOLUTION

One App. Every Farmer's Personal Agronomist.

Kheti Mitra is a multilingual, voice-first Decision Support System - combining LSTM yield forecasting, live government data fusion, and a Sustainability Support to give every farmer hyper-personalized, actionable advice.



AI Crop Advisor

LSTM model forecasts yield & recommends top 3 crops with profit projections



Sustainability Support

Scoring sustainability, suggesting pre-defined techniques in agriculture



Policy Gap Detection

Automatically maps farmer profile to eligible government schemes



Voice-First Multilingual

Hindi + English with USSD/IVR fallback - works without smartphones

Patent-pending sustainability framework • Pilot-ready for Haryana • Deployable on ₹1,000 Android devices

Built to Scale from Day One



FRONTEND

React.js • Recharts • Tailwind CSS • Hindi/English i18n



BACKEND / API

FastAPI (Python) • Uvicorn • SQLite • REST endpoints



ML / AI

LSTM (PyTorch) • Scikit-learn • IndicTrans NLP • 84% accuracy



DATA SOURCES

eNAM Mandi API • OpenWeatherMap • NSSO • Soil Health Card



INFRASTRUCTURE

Vercel (frontend) • Render.com (backend) • GitHub

Meet Ramesh - A 2-Acre Farmer in Karnal, Haryana



RAMESH

Tenant farmer, 2 acres

Karnal district, Haryana

Annual income: ₹1.2L

No agronomist access

Basic Android phone

Speaks Hindi only

01

Farmer Dashboard

Ramesh opens the app. His crop calendar, today's weather, and Karnal mandi prices load instantly.

02

AI Crop Advisory

He inputs land size, soil type & water source. KhetiMitra recommends Top 3 crops with LSTM confidence scores.

03

Sustainability Score

Shows score based on pre-defined sustainability techniques. He sees where he can improve to increase income.

04

Farming Technique & Policy Alert

System detects Ramesh qualifies for PM-KISAN & PMFBY. Direct application links shown. He claims ₹6,000 in 2 clicks.

Real Data. Real Farmers. Real Change.

84%

LSTM Accuracy

Kharif season backtest

6

Data Sources

Govt & open APIs fused

5

SDG Dimensions

Sustainability framework

15

States Scalable

After Haryana pilot

Validation

- 10 years of Haryana crop-yield data (NSSO)
- eNAM live mandi price integration
- PM-KISAN, PMFBY, PMKSY policy logic
- 7 farmer profiles modelled & tested

Roadmap

- Phase 1: Haryana pilot — 500 farmers
- Phase 2: Punjab + UP — 50,000 users
- Phase 3: National rollout — 15 states
- B2G licensing to state agri departments

Three Revenue Streams, One Mission

B2G



Government Licensing

Sell to state agriculture departments as an AI policy tool for farmer welfare programs and scheme monitoring.

B2B



Agri-Input Companies

License advisory engine to fertilizer, seed, and equipment companies for hyper-targeted farmer outreach.

B2C



Freemium App

Free advisory + premium soil analysis, full sustainability reports, and personalized agronomy consultations.

Competitive Moat:

Unlike Kisan Suvidha (information portal), KhetiMitra is a personalized decision engine - specific to YOUR land, YOUR soil, YOUR crop history. With multilingual voice chat and agent call based system.

"Every year, Indian farmers lose ₹63,000 crore - not because the land is bad, but because the decision is wrong. Wrong crop. Wrong time. Wrong input."

WE ARE BUILDING THE MISSING LAYER

Between India's 140 million farmers and the digital economy.

Voice-First

No literacy barrier

High Impact on lives

Farmers & Industries

Pilot-Ready

Haryana → 15 States

